Modern Search and Sorting Benchmark

- Individual C++17/Python research project
- Correctness first; reproducible raw data; scope-specific observations
- CPU experiments complete for delivered matrices; frozen CPU/GPU full study remains blocked

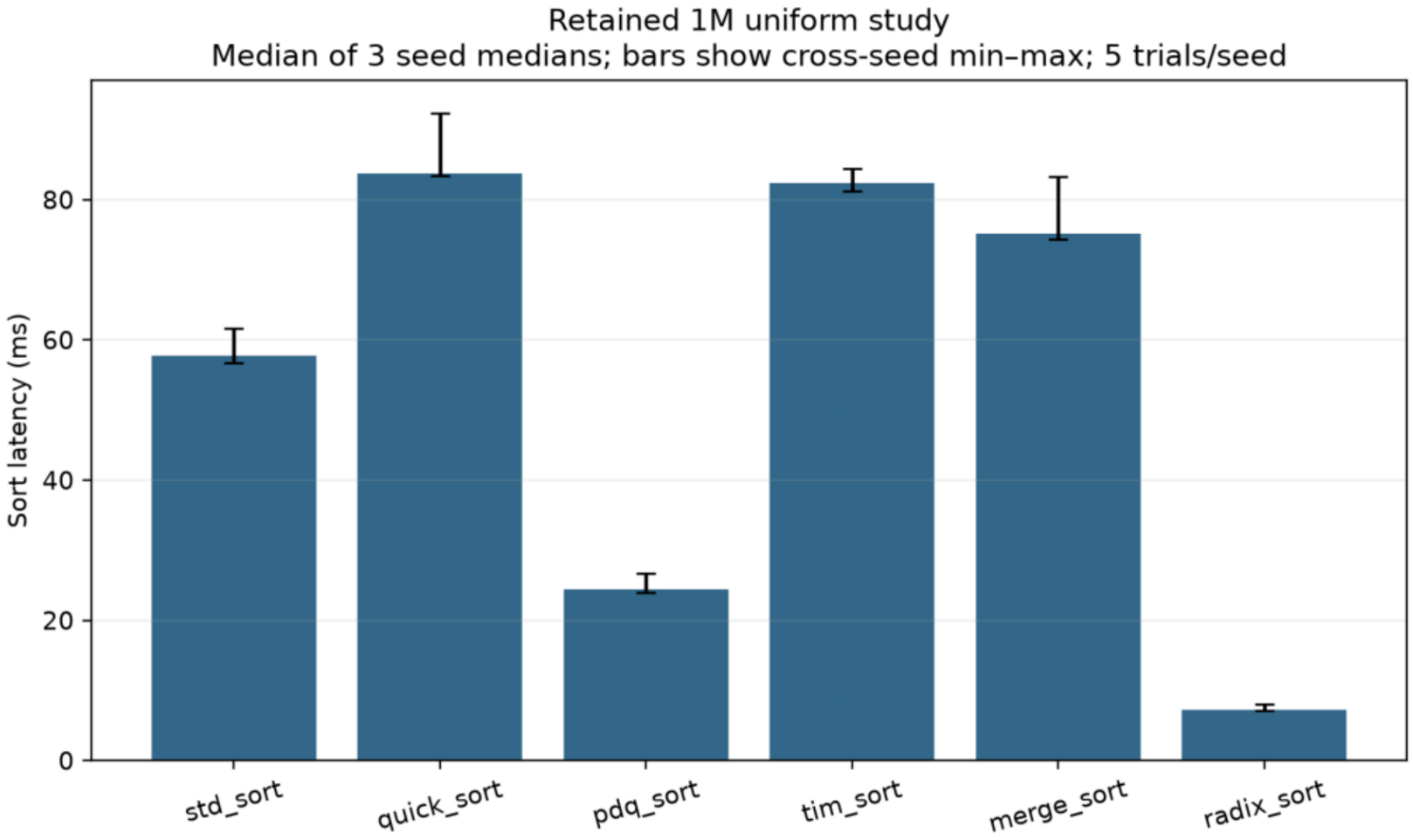
Research questions and implementations

- Six sequential sorters: std_sort, QuickSort, PDQSort, Timsort, Merge Sort, Signed Radix Sort
- Linear/Binary/Exponential Search; Skip List; B-tree orders10/16/32/64
- OpenMP Merge Sort; CUB radix adapter; endpoint learned rank index

Experimental design

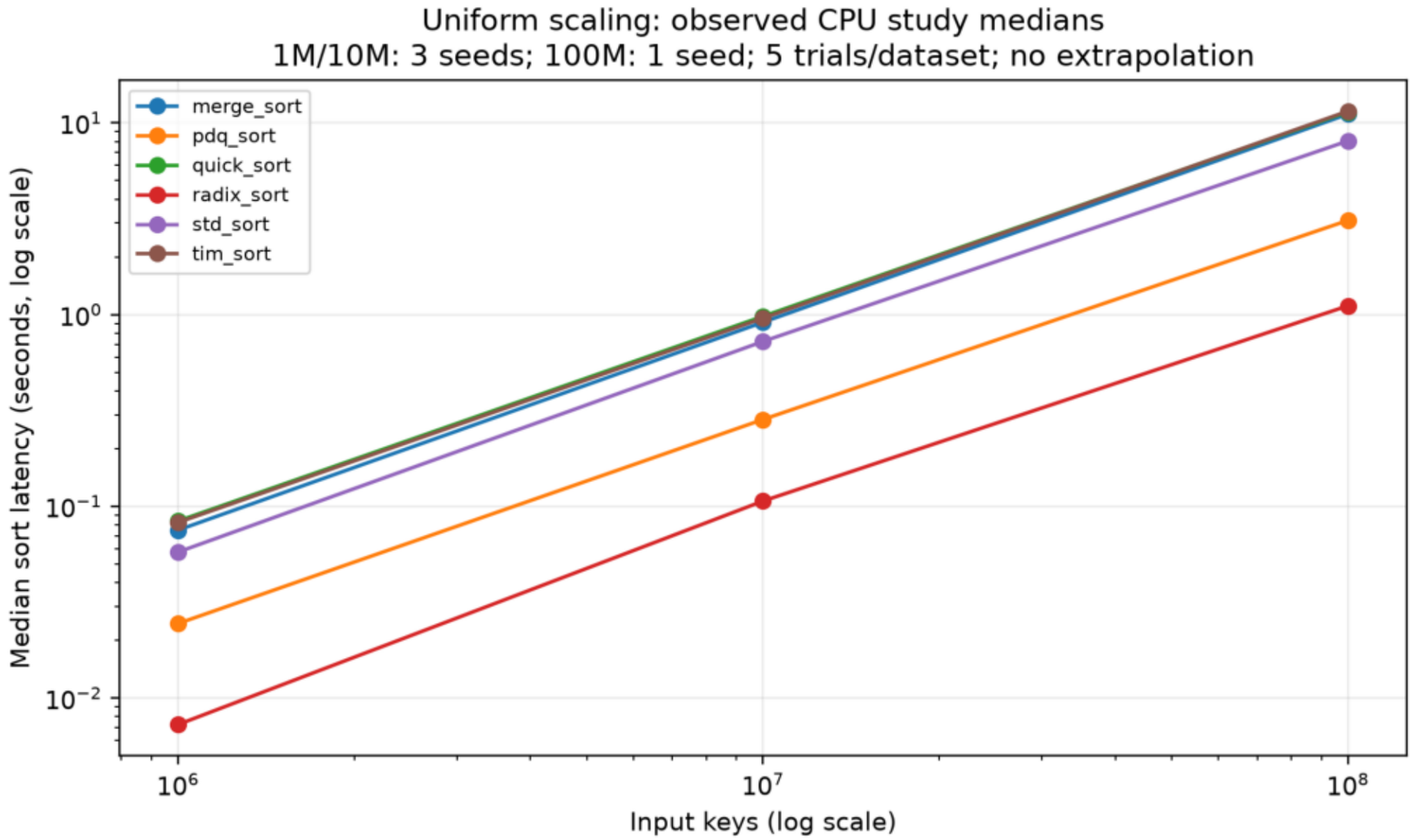
- Deterministic signed-int32 generator; identical inputs; oracle equality outside timing
- Release, steady_clock; one warm-up/five measured repetitions
- Separate counts/RSS and GPU event scopes; no hidden setup or device-only speedup ratio
- Dataset medians first; no outlier removal or unsupported significance claims

1M sorting: workload-specific observations



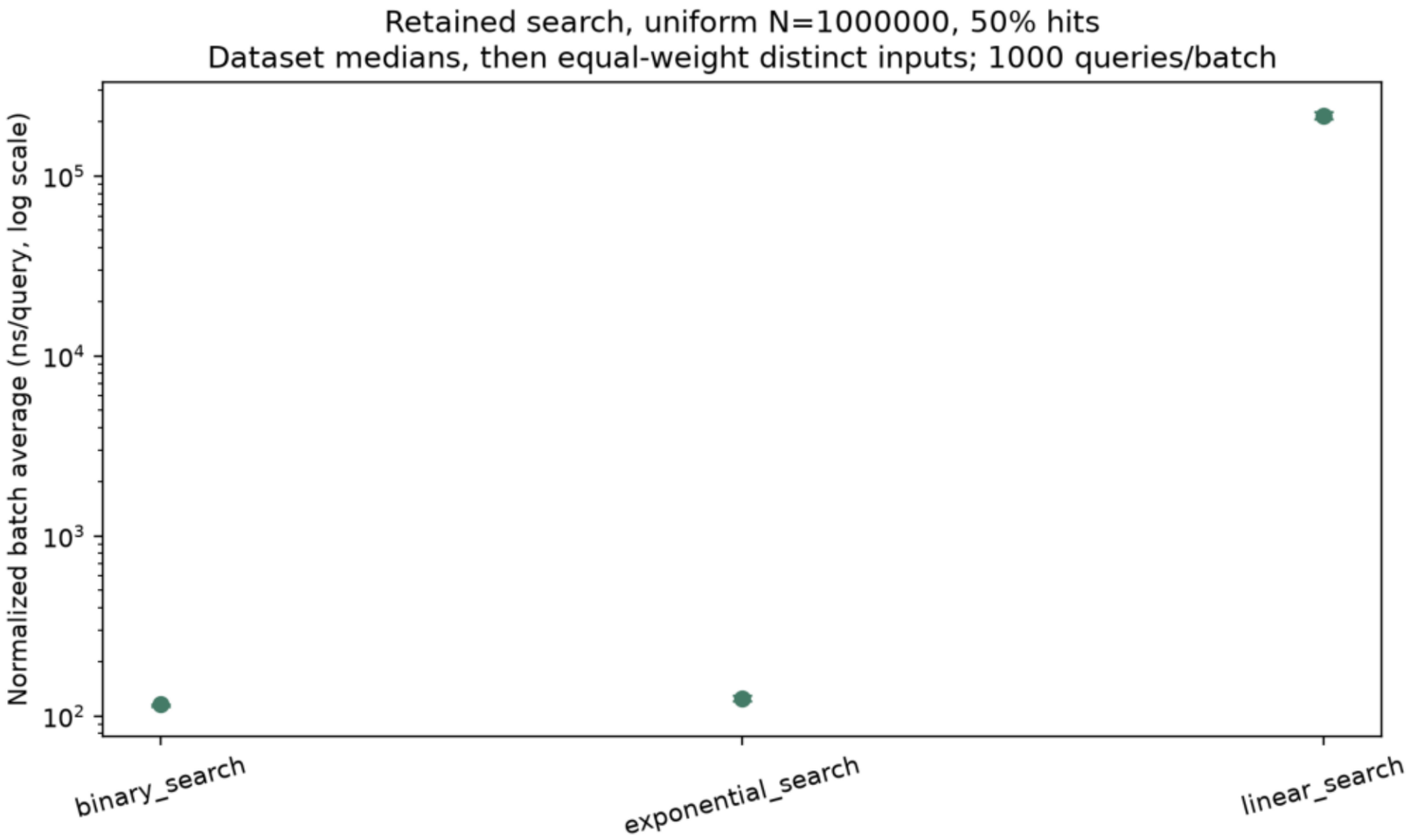
- Uniform group shown; other distributions differ
- Three seed datasets, five trials each; spread is not a confidence interval

CPU scaling through100M



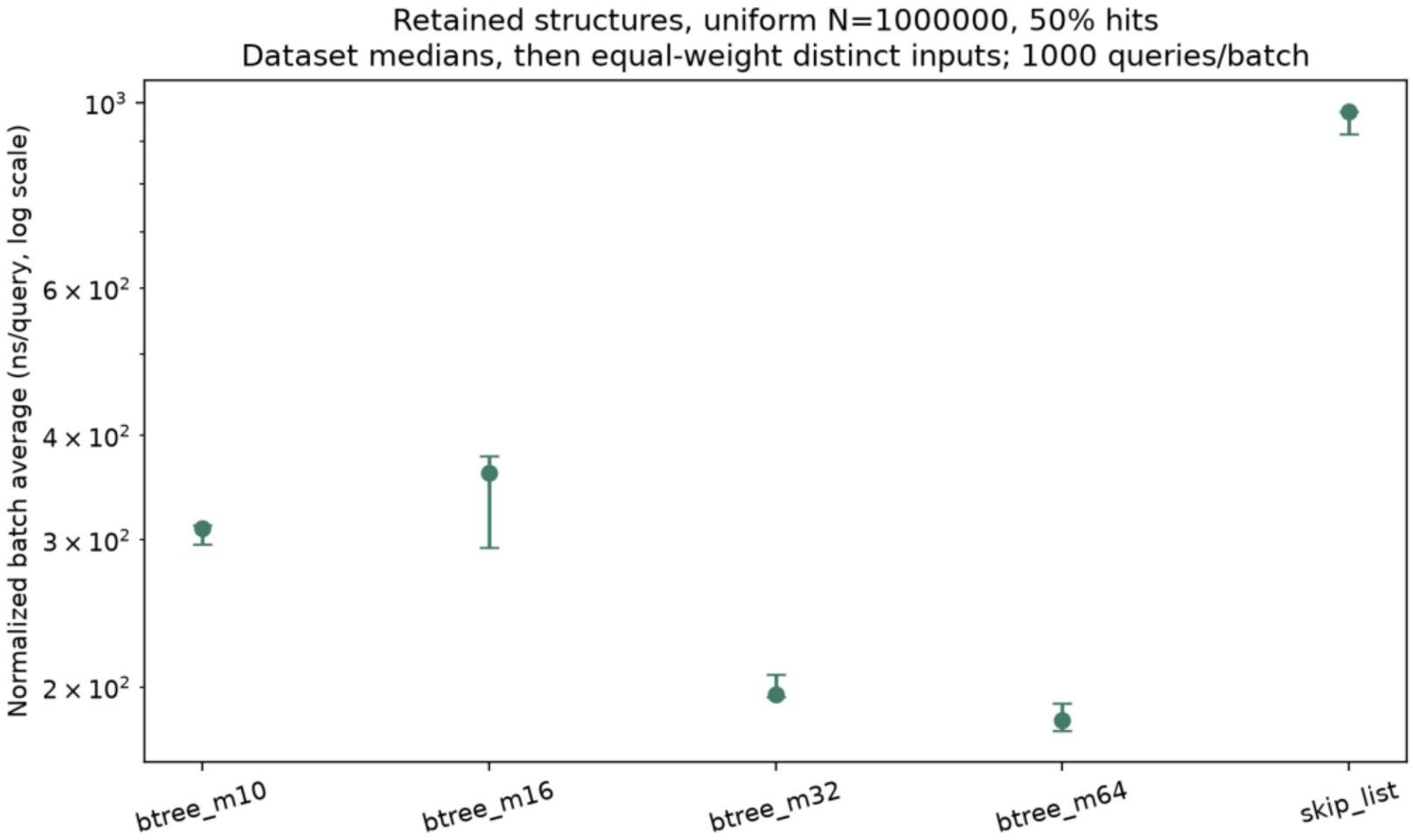
- 10M:450 measured +90 warm-ups;100M:150 measured +30 warm-ups
- All six sorters, five distributions;100M has one seed
- Log/log axes; distinct study sessions; no extrapolated performance

Search: query outcome and position matter



- Same1000-query mixed50% batch for each searcher
- Normalized batch average, not directly measured individual-query latency

Ordered structures



- Skip List and B-tree variants; unique-key set contract

- Construction and lookup timed separately; representative uniform lookup shown

Learned rank index

- Endpoint affine prediction; radius32 correction; certified binary fallback
- 48 held-out configurations;480 measured lookup batches;240 measured builds
- 48000 separate diagnostics; counts, correction distance, fallback and logical storage retained
- No constant-time guarantee; duplicate-heavy data can be poorly predicted

Parallel and GPU findings

- Historical OpenMP sweep:1800 measured sorts; sequential Merge Sort is the primary baseline
- Fresh untimed worker diagnostic confirms team threshold, not task participation/race freedom
- T4 correctness and prepared timing smoke are historical, validated separately
- Full frozen CPU/GPU comparison NOT launched: external trust anchor and confirmed free runtime unavailable

Comparison counts and memory

- 240 separate instrumented executions; five comparison adapters; Radix count unsupported
- Comparison counts exclude moves, bookkeeping and instructions
- RSS is whole-process lifetime peak, not isolated auxiliary allocation
- All plots derive from retained observations; missing metrics remain explicit

Limitations and AlphaDev

- Background load, frequency, thermals and cache state not controlled
- Three seeds do not establish statistical significance;100M has one seed
- Batch averages cannot establish maximum individual-query latency
- AlphaDev primary-source review only: no training/discovery reproduction

Conclusion and reproducibility

- CPU core, learned evaluation, scaling and this report are reproducible from retained data
- Historical evidence and frozen protocols preserved; failed attempts retained
- Frozen CPU/GPU launch guards remain enforced; no fabricated approval or provenance
- INPUTS_SHA256.json, PLOTS.json, audit and command records identify every derived claim